

Functional sequences, series and spaces – 1

Sequences of functions and uniform convergence

Homework

13 сентября 2026 года

Theoretical minimum. Uniform convergence is the statement $\sup |f_n - f| \rightarrow 0$ and is checked either directly or through the Cauchy condition. It preserves continuity and the Riemann integral, and it does not preserve differentiability: there the hypothesis must be put on the derivatives. Monotonicity upgrades pointwise convergence to uniform in two ways, through Dini and through Pólya. The only general way to extract a convergent subsequence from a family of functions is Arzelà–Ascoli, that is, uniform boundedness plus equicontinuity; on a non-compact domain one gets uniform convergence on compact subsets and nothing more. Two reflexes worth having: to disprove uniformity, find a moving point x_n where $|f_n(x_n) - f(x_n)|$ stays away from zero; and to prove it, estimate the supremum honestly rather than each value separately.

1. Prove that a uniform limit of Riemann integrable functions is integrable and that the integrals converge. Explain why the analogous statement for pointwise convergence fails, and compare with the theorem of Arzelà in Definite integrals – 8.
2. Let $f_n \in C^1[0, 1]$ converge uniformly to f . Give an example where f is nowhere differentiable, and prove that if in addition f'_n converges uniformly, then $f \in C^1$ and $f' = \lim f'_n$.
3. (Mexmat, 2015) A sequence of functions f_n on $[0, 1]$ is given. Is it true that one can always find numbers $\varepsilon_n > 0$ such that $\lim_{n \rightarrow \infty} \varepsilon_n f_n(x) = 0$ for every $x \in [0, 1]$?

4. (Mexmat, 2001) Let (α_n) be a sequence of real numbers such that the sequence of functions $\sin(\alpha_n x)$ converges on a set of positive measure. Prove that (α_n) has a finite limit.
5. Let (f_n) be equicontinuous on $[a, b]$ and suppose $f_n(x)$ converges for every x in a dense subset. Prove that (f_n) converges uniformly on $[a, b]$.
6. Prove that the closed unit ball of $C[0, 1]$ is not compact by exhibiting a uniformly bounded sequence with no pointwise convergent subsequence, and prove that your example indeed has none.
7. Let $f_n \rightarrow f$ uniformly on (a, b) and suppose each f_n has a limit L_n as $x \rightarrow a^+$. Prove that (L_n) converges and that $\lim_{x \rightarrow a^+} f(x) = \lim L_n$; give an example showing that pointwise convergence is not enough.
8. Construct continuous functions f_n on $[0, 1]$ with $f_n \rightarrow 0$ pointwise and $\int_0^1 f_n \rightarrow 0$ for which $\max f_n \rightarrow \infty$, and prove that such a sequence cannot be monotone in n .
9. Prove that a uniform limit of polynomials of degree at most d is a polynomial of degree at most d , and give an example of a uniform limit of polynomials of unbounded degree which is nowhere differentiable.
10. Let $f_n: [0, 1] \rightarrow [0, 1]$ be continuous and suppose $f_n \rightarrow f$ pointwise with f continuous. Must the convergence be uniform on some subinterval? Prove it or construct a counterexample.